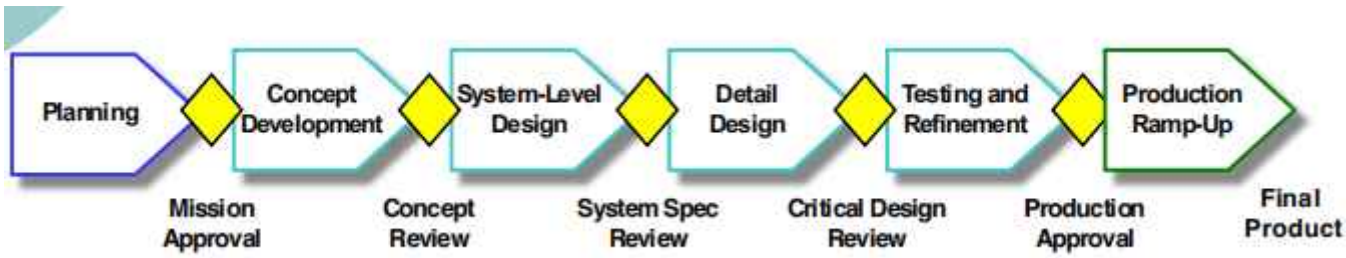
产开

> A generic product development process
>
> ¡ The process has six distinct phases 产品开发的六步

总流程：



Topic 2 – Project Management

Project Management is planning and coordinating resources and tasks for降本增质，提高效率

Two phases:

- Project Planning：scheduling the project tasks and determining resource requirements. 安排任务 资源需求
- Project Execution：complete the project in the face of inevitable unanticipated events and the arrival of new information. 确定目标 坚决执行

Understanding and Representing Tasks：

- Sequential, Parallel, and Coupled Tasks 顺序并行耦合
- The Design Structure Matrix 结构矩阵
- Gantt Chart 甘特图
- PERT Charts 图表
- The Critical Path 关键路径

Baseline Project Planning：

A roadmap for the remaining development effort.规划资源和时间 越早越重要

创建的方法：

- Contract Book 合同手册
- Project Task List 任务清单
- Team staffing & Organisation 人员组织
- Project Schedule 项目规划 merger of the project tasks and the project time line 任务+时间线
- Project Budget 项目预算
- Project Risk Plan 风险规划
- Modifying the baseline plan 修改基准计划

4个步骤：

1. identify the dependencies among tasks. DSM矩阵确定依赖
2. Position the key project milestones along a time line in a Gantt chart.甘特图做时间表
3. Schedule the tasks (staffing and resources)
4. Adjust the timing of the milestones to be consistent with the time required for the tasks. 2和3协调一下

Accelerating Projects

总的看：

- Start the project early 早点开始
- Manage the project scope 时间范围
- Facilitate the exchange of essential information 信息交换

关键路径上看decreasing the time：

- Complete individual tasks on the critical path more quickly 每个都快点
- Aggregate safety times 聚合安全时间
- Eliminate some critical path tasks entirely 削任务
- Eliminate waiting delays for critical path resources 减少等待时间
- Overlap selected critical tasks 重叠关键任务
- Pipeline large tasks 排查大型任务

- Outsource some tasks 外包

任务加速at completing coupled tasks more quickly：

- Perform more iterations quickly 快速迭代
- Decouple tasks to avoid iterations 解耦 避免迭代
- Consider set of solutions 多考虑解决方案

Coordination Mechanisms协调机制

- Informal communications
- Meetings
- Schedule display
- Weekly display
- Incentives
- Process documents

Summary

- Project management is the activity of planning and coordinating resources and tasks to achieve high-quality, low-cost products while making efficient use of time, money, and other resources.
- Project planning involves scheduling the project tasks and determining resource requirements.
- Project execution, sometimes called project control, involves coordinating and facilitating the myriad tasks required to complete the project in the face of inevitable unanticipated events and the arrival of new information.

l 0. Planning - Topic 4 - Product Planning

Product Planning mean

before a product **development** project is formally **approved**

before substantial **resources** are applied

before the larger development **team** is formed

Product Planning Problems

l Inadequate **coverage** of target **markets** with competitive products

l Poor **timing** of market **introductions** of products

l **Mismatches** between aggregate development **capacity** and the **number** of projects pursued

l Poor **distribution** of resources, with some projects **overstaffed** and others **understaffed**

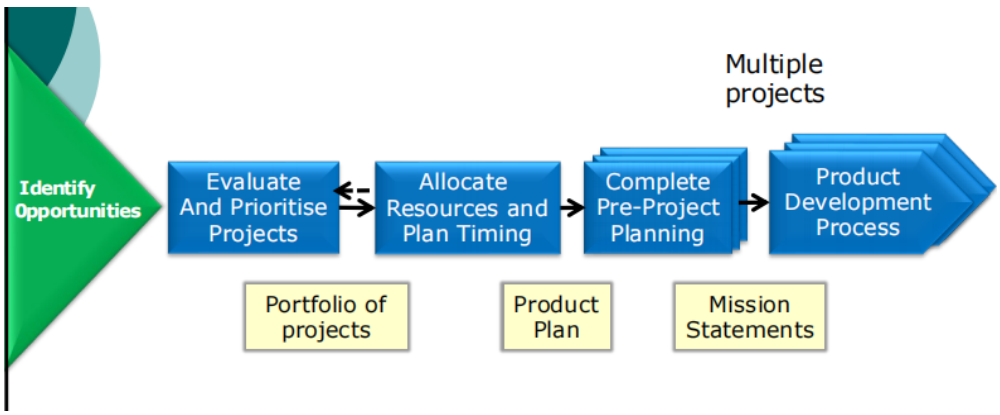
l Initiation and subsequent cancellation **of illconceived** projects

l Frequent **changes** in the directions of projects

Four types of Product Development Projects

1. **Fundamentally new products**
2. **New Product *Platforms***
3. **Derivatives of existing product *platforms***
4. ***Incremental improvements to existing products***

5 steps product planning process



Step 1 – Identify Opportunities

见topic3

Step 2 – Evaluate and prioritise projects

Decide **which** of the projects to pursue by **four perspectives**

l Competitive strategy

l Market segmentation

l Technological trajectories

l Product platforms

Step 3 – Allocate resources and plan timing

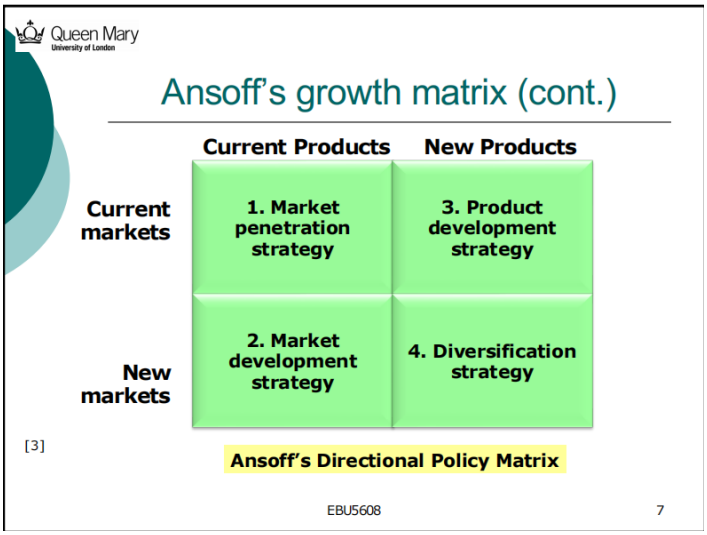
l Resource **Allocation**

l Project **Timing**

Step 4 – Complete pre-project planning

Step 5 – Reflect on the results and the process

l Step 1 – Topic 3 Identify Opportunities



Effective opportunity tournaments

Opportunities Funne

Opportunity identification process

¡ Establish **a charter**

articulates the **goals of the organisation** and establishes the **boundary conditions for an innovation** effort.

¡ Generate and sense **many opportunities**

Focus has to be both on **internal** and **external sources** of raw opportunities.

1. Internally e.g. **R&D department**
2. By the current or potential **customer**
3. Through a **competitive product**
4. By **sales forces**
5. Through collaboration with **universities**
6. By independent **inventors**
7. By **distribution partners**
8. By **other partner** companies

Where do opportunities come from?**passively proactively.**

Role of R&D

Research

Development

l R&D for **existing businesses**

l R&D for **new businesses**

l R&D for **exploratory research**

Corporate Strategy-> business strategies

¡ **Screen** opportunities

The purpose of screening is

l to **eliminate** any opportunities that are **unlikely to result in the creation of value**,

l to **focus** attention on the **opportunities worthy** of further investigation

l **not to pick the single best opportunity!**

There are two effective **approaches to screen**

opportunities:

l Web-based **surveys**

l Workshops with **multi-voting**

¡ Develop **promising opportunities**

invest **modest levels of resources in developing** a few of the screened opportunities.

l **customer interviews**,

l **testing** of existing products,

l **concept generation**,

l quick **prototypes**,

l estimates of **market sizes** and **growth** rates.

¡ Select **exceptional opportunities**

select a few that warrant a **significant investment** in product development.

*3 **Real-Win-Worth-it:***

l Is the opportunity **real?**

l Can you **win** with this opportunity?

l Is the opportunity **worth it** financially?

¡ **Reflect** on the result and process.

l How **many of the opportunities** identified came from **internal** sources versus **external** sources?

l Did we consider **dozens or hundreds** of opportunities?

l Was the **innovation charter** too narrowly **focused**?

l Were our **filtering criteria biased**, or largely based on the best possible estimates of eventual product success?

l Are the resulting opportunities **exciting to the team**?

Summary

¡ Opportunity, definition and types

¡ Opportunity identification process

¡ Various steps in opportunity identification

¡ Role of Research and Development

l Step 2 – Evaluate and prioritise projects

见**Topic 4 - Product Planning**

l Step 3 – Allocate resources and plan timing

见**Topic 4 - Product Planning**

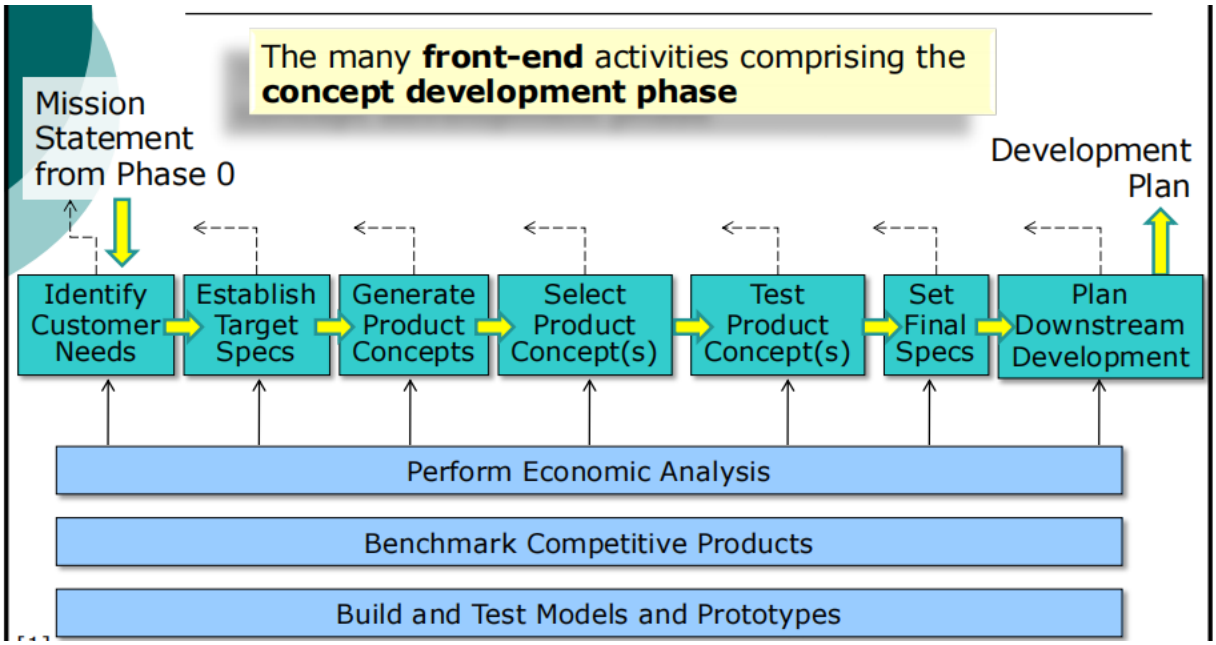
l Step 4 – Complete pre-project planning

见**Topic 4 - Product Planning**

l Step 5 – Reflect on the results and the process

见**Topic 4 - Product Planning**

l 1. Concept development - Topic 6（重注）



¡ The **needs** of the target market are identified

¡ **Alternative** product concepts are generated and evaluated

l A **concept** is a **description** of the form, function and features of a product

¡ One or more concepts are selected for further **development** and **testing**

¡ **Evaluation** and **screening criteria** are used to aid in the **selection**

¡ Usually accompanied by a set of **specifications**, an **analysis** of competitive products and an economic **justification** for the project

l Identify Customer Needs

Customer need statements organised in a hierarchical list, with **importance** weightings for many or all of the needs

Market Research

l the function that links the consumer, customer and public to the marketer through **information** –information used

¡ to identify and define marketing **opportunities** and **problems**;

¡ to generate, refine and evaluate marketing **actions**;

¡ to monitor marketing **performance**;

¡ to improve **understanding** of the marketing process

5 steps:

1. Gather raw data from customers

Collection of **primary/secondary data**

¡ Data may be qualitative and/or quantitative

¡ Marketing studies often start with a **review** of **secondary** data. This can form a basis for **designing** and **carrying** out the more difficult and expensive **primary** data collection

Concept Research

l Develop decision making **strategies**

l Examine **existing** market segments

l Understand shopping **habits**

l Understand the **customers' view** of the

products available

l Look at customers' **beliefs/expectations**

l Analyse current **buying patterns**

l Carry out a **SWOT/PESTEL** analysis

2. Interpret the raw data in terms of customer needs

¡ Date be used to express the customers' needs in terms of what the product has to do

¡ Use **positive**, not **negative** phrasing

¡ Express the needs as **attributes** of the product

3. Organize the needs into a hierarchy of needs

¡ **Structure** the needs into

- **Must-haves** – "*I wont buy without*"
- **Delighters** – "*What an unexpected treat*"
- **Linear Satisfiers** – "*The more the merrier*"
- **Neutrals** – "*No big deal*"

¡ This is the **Kano** classification

¡ Consumer needs can be very **elusive**

¡ **Intuitions** are often wrong

4. Establish the relative importance of the needs

Measuring **preference** is central to market research

5. Reflect on the results and the process

No process is an **exact science**

Key Benefits

¡ Ensuring that the product is focused on customer needs and that no critical customer need is forgotten;

¡ Developing a clear understanding among members of the development team of the needs of the customers in the target market;

¡ Developing a fact base to be used in generating concepts, selecting a product concept, and establishing product specifications;

¡ Creating an archival record of the needs phase of the development process.

l Establishing target specifications

¡ **Specifications** provide a precise **description** of what a product has to do

¡ Are a **translation** of the customer needs into **technical** terms

¡ The **output** of this stage is:

l A list of target **specifications**

l Each specification consists of a **metric**, and marginal and ideal **values** for that metric

	Need \ Metric	1 Attenuation from dropout to handlebar at 10hz	2 Spring preload	3 Maximum value from the Monster	4 Minimum descent time on test track	5 Damping coefficient adjustment range	6 Maximum travel (26in wheel)	7 Rake offset	8 Lateral stiffness at the tip	9 Total mass	10 Lateral stiffness at brake pivots	11 Headset sizes	12 Steerube length	13 Wheel sizes	14 Maximum tire width	15 Time to assemble to frame	16 Fender compatibility	17 Instills pride	18 Unit manufacturing cost	19 Time in spray chamber w/o water entry	20 Cycles in mud chamber w/o contamination	21 Time to disassemble/assemble for maintenance	22 Special tools required for maintenance	23 UV test duration to degrade rubber parts	24 Monster cycles to failure	25 Japan Industrial Standards test	26 Bending strength (frontal loading)
1	reduces vibration to the hands.	•			•																						
2	allows easy traversal of slow, difficult terrain.	•																									
3	enables high speed descents on bumpy trails.	•			•																						
4	allows sensitivity adjustment.					•																					
5	preserves the steering characteristics of the bike.						•	•																			
6	remains rigid during hard cornering.		•						•																		
7	is light weight.									•																	
8	provides stiff mounting points for the brakes.										•																
9	fits a wide variety of bikes, wheels, and tires.											•	•	•	•												
10	is easy to install.															●											
11	works with fenders.																•										
12	instills pride.																	•									
13	is affordable for an amateur enthusiast.																		•								
14	is not contaminated by water.																			•							
15	is not contaminated by grunge.																					•					
16	can be easily accessed for maintenance.																						•				
17	allows easy replacement of worn parts.																						•	•			
18	can be maintained with readily available tools.																							•			
19	lasts a long time.																								•	•	
20	is safe in a crash.																									•	•

4 steps to this process:

1. Prepare the list of **metrics**
2. Collect competitive benchmarking information
3. Set ideal and marginally acceptable target values
4. Reflect on the results and the process

l Concept generation

to thoroughly **explore** the space of product concepts that may address the customer needs

include:

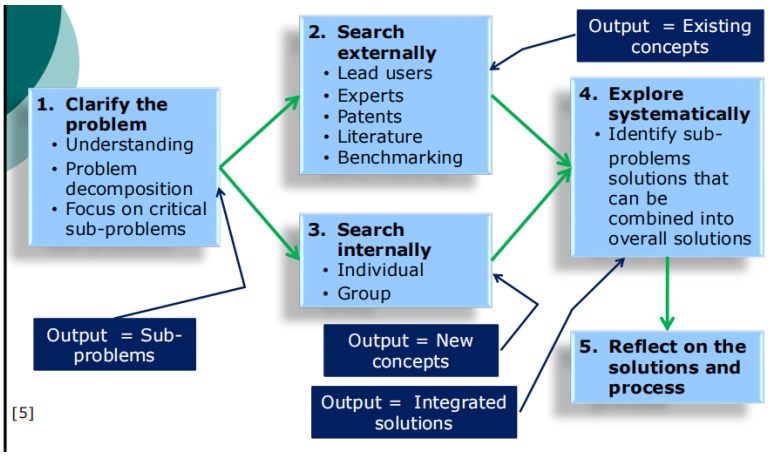
l external **search**

l creative **problem solving** within the team, and

l systematic exploration of the various **solution fragments the team generates**

Creating by:

1. break the problem into **simpler sub-problems**
2. sub-problem solutions are **integrated** into a total solution



l Concept selection

Various product concepts are **analysed** and sequentially **eliminated** to identify the most **promising** concept

Usually requires several **iterations** of the screening process

Screening and evaluation of business opportunities

5 stages：

1. **Initial screen**

First **formal evaluation** of the idea come from a pool of concepts

Screening will include：

Technical feasibility check、**Marketing feasibility** test、Evaluation of the **strategic opportunity**

未来决定的依据，错了以后损失大

Evaluation criteria can be used to **formalise** the decision-making process regarding project funding and use **scoring model** or **weighted checklist**

Criteria such as **Technical、Research direction and balance、Competitive rationale、Patentability、Stability of the market、Integration and synergy、Market、Channel fit、Manufacturing、Financial、Strategic fit**

2. **Customer** screen

informal discussion with **customers** to explain a concept

a **prototype** is a useful discussion and explanation tool, especially where customers are **non technical** or are just **end-users**

3. **Technical** screen

informal technical **discussions** with experts、extensive **analysis** by a 3rd party

4. **Final** screen

Involves the use of screening models and computer assessment programs

5. **Business** analysis

l preliminary **marketing** plans,

l **technical** plans,

l **financial** reviews and

l projected **budgets**

Potentially **new problems** are identified

l Concept testing

l verify that the **customer needs** have been met

l assess the **market potential** of the product and any **shortcomings** that need further development

7 steps：

a. **Define the purpose of the concept test**
b. Choose a survey population
c. Choose a survey format
d. Communicate the concept
e. Measure customer response
f. Interpret the results
g. Reflect on the results and process

l Setting final specifications

Target **specifications** are **revisited**

The team must commit to specific values of the **metrics** reflecting

l the **constraints inherent** in the product concept

l **limitations** identified through technical **modelling** and

l **trade-offs** between cost and performance

l Project planning

The team

l creates a detailed development **schedule**

l devises a strategy to **minimise** development time and

l identifies the **resources** required to complete the project

Contract book

l Mission statement

l Customer needs

l Details of the selected concept

l The product specifications

l The economic analysis of the product

l The development schedule

l The project staffing

l The budget

l Economic analysis

l justify **continuation** of the overall development programme and

l resolve specific **trade-offs** among **development** costs and **manufacturing** costs

l Benchmarking & modelling

Benchmarking of competitive products

l An understanding of **competitive** products

l Can be a rich source of **ideas** for the product and production process design

Modelling and **prototyping**

l Every stage of **development** process involves various forms of models and prototypes

l 2. System-level design

Phase 2 includes the definition of the **product architecture** and the **decomposition** into functional & physical **elements.**

1)Product Architecture

Description:

l 1. “the **scheme** by which the **functional elements** of the product are **arranged** into **physical chunks** and by which the **chunks interact**”

l 2.**Functional** – individual **operations** and **transformations** that contribute to its **overall** performance

l 3.**Physical** – **parts**, **components** and **subassemblies** that implement the product’ s functions

Aim

Architectural decisions allow the **detailed design** and **testing** of these physical blocks to be assigned to **teams**,individuals, and/or suppliers, so that the development of different portions can be carried out **simultaneously.**

Chunk

1)The physical elements of a product are typically organised into several major building blocks – called **chunks**

2)These chunks become the building blocks for the product or **family** of products

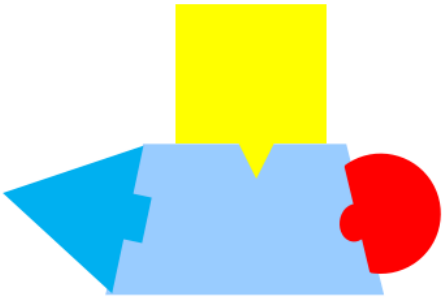
3)Each chunk is then made up of a collection of **components** which implement the **functions** of the product

2)Modular Architecture

There are three main types of modular architecture:

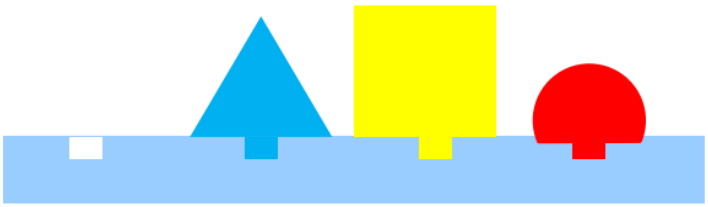
Slot-modular

Each of the **interfaces** between chunks in a slot-modular architecture is of a **different type** from the others – therefore, the various **chunks** in the product **can not** be interchanged



Bus-modular

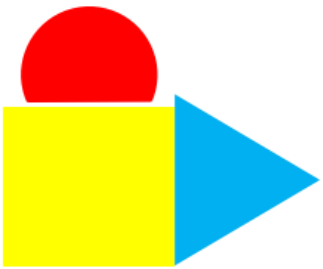
There is a **common bus** to which the other chunks connect via the **same type** of interface



Sectional-modular

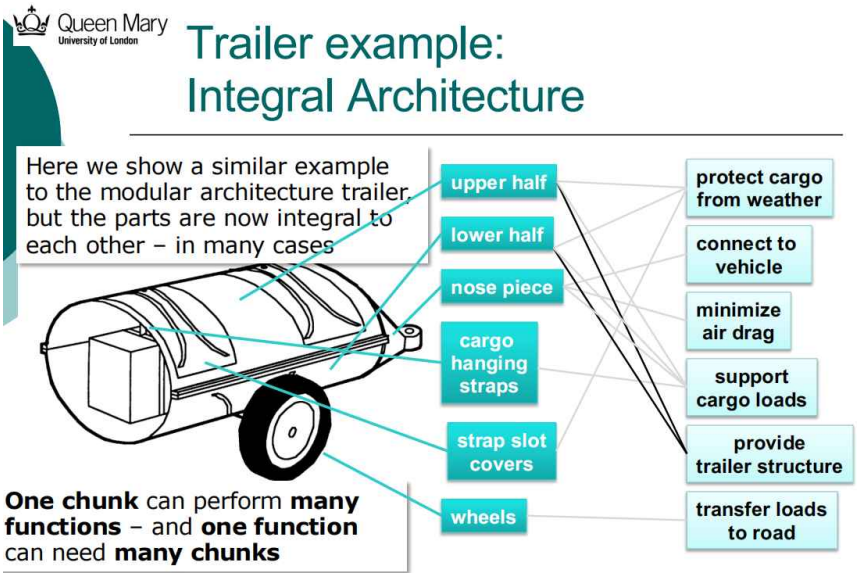
i) All **interfaces** are of the **same type**. There is **no single element** to which all the other chunks **attach**

ii) The assembly is built up by connecting the chunks to **each other** via identical interfaces



3)Integrity Architecture

Example



4)Implications of product architecture

There are **implications** to the decisions you make about product architecture

Product change

l **Modular** chunks allow **changes** to be made to a **few isolated** functional elements of the product

without necessarily affecting the design of **other chunks;**

l **Motives** for change include – upgrades, add-ons, adaptation, wear, consumption, flexibility in use,

Reuse.

Product variety

1）Variety refers to the **range** of product models the firm can produce within a particular **time** period in response to market **demand**

2）Products built around **modular** product architectures can be **more easily varied** without adding tremendous complexity to the **manufacturing** system

e.g. mobile phone handset design, portable audio

Example：

Sony is a major consumer and commercial electronics manufacturer

l Sony sells **54 models** of portable audio equipment in the UK, including

- l MP3
- l Personal CD, minidisc & tape
- l Portable CD+radio

l Portable radio

Uses **modular architecture** and **flexible manufacturing** techniques to

¡ Introduce new **models**

¡ Meet changes in **demand**

Component standardisation

i ） The use of the **same** component or chunk in **multiple** products

ii ） If a chunk implements only one or a few **widely useful** functional elements, then the chunk can

be **standardised** and used in several **different** products

iii） Example – standard batteries

Product performance

1) How **well** a product implements its intended **functions**

2) Product performance **characteristics** include speed, efficiency, life, accuracy and noise

Manufacturability

The product architecture also directly affects the ability of the team to design each chunk to be

produced at **low cost.**

Product development management

1) **Responsibility** for the detail design of each chunk is usually assigned to a relatively **small group** within the **firm** or to an outside **supplier**

2) Chunks are assigned to a single individual or group because their design requires careful resolution of **interactions** among components within the chunk

5)The 4 step method

Ulrich and Eppinger recommend a **4-step method**

for establishing the product architecture

l Step 1 – Create a **schematic** of the product

1.representing understanding of the **component elements** of the product.

2.The schematic should reflect the **best understanding** of the state of the product, it does not have to contain every detail

3. More **detailed** functional elements are finalised **later** in Phase 2

4. A good **guideline** is for there to be **no more than 30 elements** in the schematic

5. Usually **more than one** alternative schematic is developed and the team spend time **selecting** the

most appropriate.

l Step 2 – **Cluster** the elements of the schematic

1.This step requires **assigning** and/or **clustering** each of the **elements** of the schematic into a **chunk**

2.Setting the clusters.The clustering role is undertaken by the **team** to ensure that the right decisions are taken in terms of **functionality split**

l Step 3 - Create a rough **geometric layout**

1.**Geometric integration and precision**

2. Elements requiring **precise location** or **close geometric integration**

3.**Function sharing:** When a **single** physical **component** can implement **several functional** elements of the product, these functional elements are best clustered **together .**

4.**Capabilities of vendors**

5.**Similarity of design or production technology**

6.**Localisation of change**

7.**Accommodating variety**

8.**Enabling standardisation**

l Step 4 – Identify the fundamental and incidental **interactions**

To **manage** this coordination process better, the team should **identify** the known **interactions** between chunks during the system-level design phase.

There are two categories of interaction

l **Fundamental**

-Those which correspond to the lines on the **schematic** that **connect** the chunks to one another

-These are the **fundamental** interactions of the systems operation

l **Incidental**

-An **incidental interaction graph** is used to document this type of interaction

Key Outcomes

The **end results** of this 4-step method for

establishing the product architecture are:

l) an approximate **geometric layout** of the product

l) **descriptions** of the major chunks

l) documentation of the key **interactions** among the chunks

l 3. Detail design （重注）

l Aim

l Control Documentation

l The **output** of Phase 3 is the **control documentation** for the product

l The control documentation is: l The **drawings** or computer **files** describing the **geometry** of each part to be made and its production **tooling**

l The process descriptions for the **fabrication** and

assembly of the product

l The specifications of the parts to be **purchased**

l Departmental responsibilities

l **Marketing**

¡ Develop marketing plan

l **Design**

¡ Define part **geometry**

¡ Choose **materials**

¡ Assign **tolerances**

¡ Complete industrial design **control documentation**

l **Manufacturing**

¡ Define piece-part **production** processes

¡ Design **tooling**

¡ Define **quality assurance** processes

l Prototype

l Definitions: Prototype, Prototyping

A prototype is

l "an **approximation** of the product along one or more dimensions of interest"

l i.e.

¡ Industrial designers produce **prototypes** of their concepts, such as models

¡ Engineers **prototype** a design

¡ Software developers write **prototype** programs

Prototyping is the process of developing such an approximation of the product.

l 3-Phase of Prototyping

¡ **Alpha** prototypes

l Are typically used to assess whether the product works **as intended**

¡ **Beta** prototypes

l Are typically used to assess **reliability** and to identify remaining **bugs** in the product

¡ **Pre-production** prototypes

l Are the first products produced by the entire **production process**

l Uses of prototypes

1. Learning

¡ Answering questions about performance or feasibility , such as

l "Will it work?"

l "How well does it meet the customer needs?"

¡ For example, a proof-of-concept model

2.Communication

l Demonstration of a product to get **feedback** from all stakeholders

¡ e.g. top management, vendors, partners, extended team members, customers and investors

l A **physical**, tactile, 3D representation of a product is much easier to understand than a verbal **description** or even a **sketch** of a product

¡ e.g. 3D physical models of style or function

3. Integration

l Prototypes are used to ensure that **components** and **subsystems** of the product **work** together as

expected

l **Comprehensive physical** prototypes are the most effective as **integration** tools in product

development projects because they require the **assembly** and physical **interconnection** of all of

the parts and subassemblies that make up a product

l The integration of the prototype forces **coordination** between different **members** of the product development team

l For example: alpha or beta test models

4. Milestones

l Provide goals for the development team' s **schedule**

l Milestone **prototypes** are defined in the product development project **plan**

¡ 'Go/no go' decisions can be at these points

l The **number** of such prototypes and their **timing** is one of the key elements of the overall **development plan**

l As a **base case**, the development team should consider using alpha, beta and pre-production

prototypes as milestones

¡ e.g. first testable hardware

l Types of prototypes

¡ Prototypes can be classified along two dimensions

¡ **Physical v analytical**

l Physical – e.g. a wooden model

¡ Physical prototypes are **tangible artefacts** created to approximate the product

¡ Aspects of the product of interest to the development team are actually **built** into an artefact for **testing** and **experimentation**

¡ **Examples** of physical prototypes include

l **models** which look and feel like the product

l **proof-of-concept** prototypes used to test an idea quickly

l experimental **hardware** used to validate the functionality of a product

l Analytical – e.g. a computer model

¡ Analytical prototypes represent the product in a non-tangible, usually **mathematical**, manner

¡ Interesting aspects of the product are **analysed**, rather than built

¡ Examples of analytical prototypes include

l computer **simulations**

l systems of **equations** encoded within a spreadsheet

l computer models of three-dimensional **geometry**

¡ Comprehensive v focused

l Comprehensive - **most** of the **attributes** of the real product

¡ Comprehensive prototypes implement **most**, if not all, of the **attributes** of a product

¡ A comprehensive prototype corresponds closely to the **everyday use** of the work prototype – i.e. it is a **full-scale**, **fully operational** version of the product

¡ An example of a comprehensive prototype is one given to customers in order to identify any remaining design flaws before committing to production

l Focussed – **one or a few attributes** of the real product

¡ Focused prototypes implement **one**, or a **few**, of the attributes of a product

¡ Examples of focused prototypes include foam models to explore the **form** of a product and wire wrapped circuit boards to investigate the **electronic performance** of product design

¡ A common practice is to use **two or more** focused prototypes together to investigate the **overall**

performance of a product

¡ By building two separate focused prototypes, the team may be able to answer its questions much earlier than if it had to create one comprehensive prototype.

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Physical vs. Analytical prototypes

Physical prototypes

- **Tangible** approximation of the product
- May exhibit **unmodelled** behaviour
- Some behaviour may be a consequence of the **approximation**
- Often best for **communication**

Analytical prototypes

- **Mathematical** model of the product
- Can only exhibit behaviour arising from **explicitly modelled** phenomena (However, behaviour cannot always be predicted)
- Some behaviour may be a consequence of the **analytical method**
- Often allow more experimental **freedom** than physical models

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Mentimeter

Focused vs. Comprehensive prototypes

Focused prototypes

- Implement **one** or a **few** attributes of the product
- Answer **specific questions** about the product design
- Generally **several** are required

Comprehensive prototypes

- Implement **many** or **all** attributes of the product
- Offer opportunities for **rigorous testing**
- Often best for **milestones** and **integration**

l Principles for choosing a Prototype type

¡ Analytical prototypes are in general more flexible than physical prototypes

¡ Physical prototypes are required to detect unanticipated phenomena

¡ Prototypes may reduce the risk of costly iterations

¡ Prototypes may expedite othe

r development steps

l Example: add a prototyping step to the part design mold design-molding process

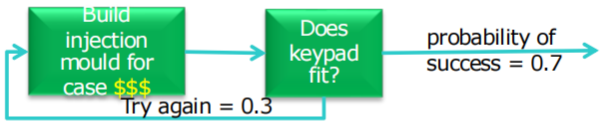
¡ A prototype may restructure task dependencies

i Prototypes may expedite other development steps

i A prototype may restructure task dependencies

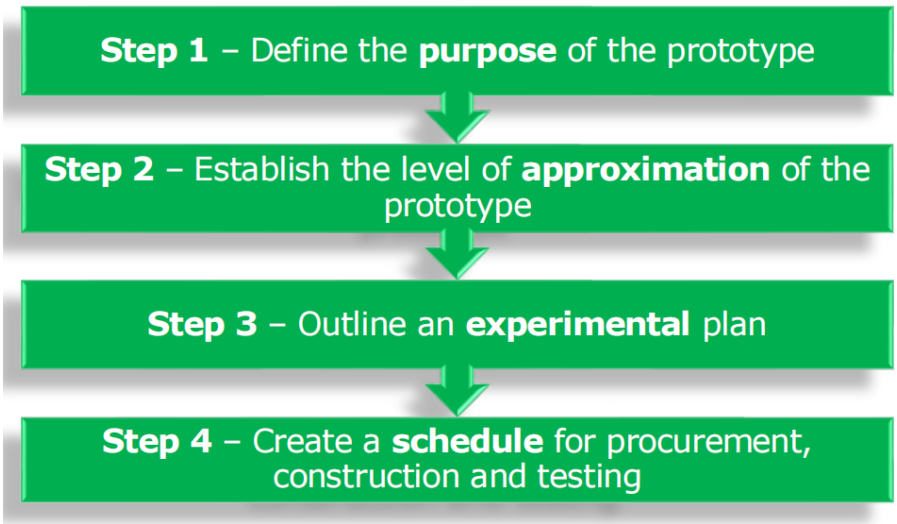
Prototypes can reduce the cost of failure during development

- *For example, does the keypad fit a plastic case for a phone?*
- **Without a prototype:**



- **With a prototype:**

l Planning a prototype



¡ Step 1 - define the **purpose** of the prototype

l The team lists its specific **learning** and **communication** needs

l Team members also

¡ list any **integration** needs

¡ decide whether or not the prototype is intended to be one of the **major milestones** of the overall product development project

¡ Step 2 – Establish the level of **approximation** of the prototype

l The degree to which the **final** product will be **approximated** in the prototype must be defined

l The team should consider whether a **physical** prototype is **necessary** or whether an **analytical** prototype would best meet its needs

l In most cases, the best prototype is the **simplest one** that will serve the purposes established in **Step 1**¡

i Step 3 – Outline an **experimental** plan

l The use of a prototype in product development can be thought of as an **experiment**

l Good **experimental practice** helps to make sure that you get the maximum value from the prototyping activity

¡ i.e. that you get useful test results

l The experimental plan includes:

¡ the identification of the **variables** of the experiment

¡ the test **protocol**

¡ an indication of what **measurements** will be performed

¡ a plan for **analysing** the resulting data

l When **many variables** have to be explored, good experimental design makes this process much easier and more effective

i Step 4 – Create a **schedule** for procurement construction and testing

l **Three dates** are particularly important in defining a prototyping effort

¡ The team defines when the parts will be ready to **assemble**

¡ The team defines the date when the prototype will **first** be tested

¡ The team defines the date when it expects to have **completed** testing and produced the **final results**

i Example of a prototype planning template

Name of Prototype	· Ball Support
Purpose (Communication, Learning, Integration, Milestones)	· What combination of spacing between ball supports and ball inertia yields the best trackball feel? (Learning) · How much variation is there in users' preferences for feel? (Learning)
Level of Approximation	· Ball surface material as planned for production design · Support material as planned for production design · Support contact geometry as planned for production design
Outline of Test Plan	· Build two sets of five different spacings for ball support · Test two balls with different inertias · Test spacings of 12.75, 13.00, 13.25, 13.50, 13.75 mm for each of the two ball inertias · Verify that all the spacings provide at least minimally acceptable performance · Have at least 20 users rank order the prototypes according to feel
Schedule	12 August Parts available 12 August Parts assembled 20 August Tests completed 22 August Analysis of results completed

l Prototyping strategy

¡ Use prototypes to reduce **uncertainty**

¡ Make models with a **defined** purpose

¡ Consider **multiple forms** of prototypes

¡ Choose the **timing** of prototype cycles

l Many **early models** are used to validate concepts

l Relatively **few comprehensive** models are necessary to test integration

¡ Plan time to **learn** from prototype cycles

l **Avoid** the "hardware swamp" - where you keep building different ways out of a problem without stopping to think

l 4. Testing and refinement

l 5. Production ramp-up

Topic 18 - Start-ups and Entrepreneurship

i Start-ups Definition

¡ A **startup company** or **start-up** is a company with a limited operating history

¡ These companies, generally newly created, are in a phase of development and research for markets

¡ Typically it involves some **novel technology, or use of technology**, but crucially…

l a coherent means for **making money** from the technology

¡ Entrepreneurship

Entrepreneur

¡ An **entrepreneur** is a person who

has possession of a new enterprise or venture and

assumes full accountability for the inherent risks and the outcome

¡ **Entrepreneurship** is the practice of starting **new** organisations or **revitalizing** mature organisations,

particularly new businesses - generally in response to identified opportunities

Two types of entrepreneurship

Social

¡ Aim is to create social change rather than make money

¡ Often involves business, public & charity organisations

¡ Examples

l poverty relief

l environment

l arts & culture

Technological

¡ Aim is to gain independence to exploit technology

¡ Often results from being frustrated in a large company

Challenges of a small business

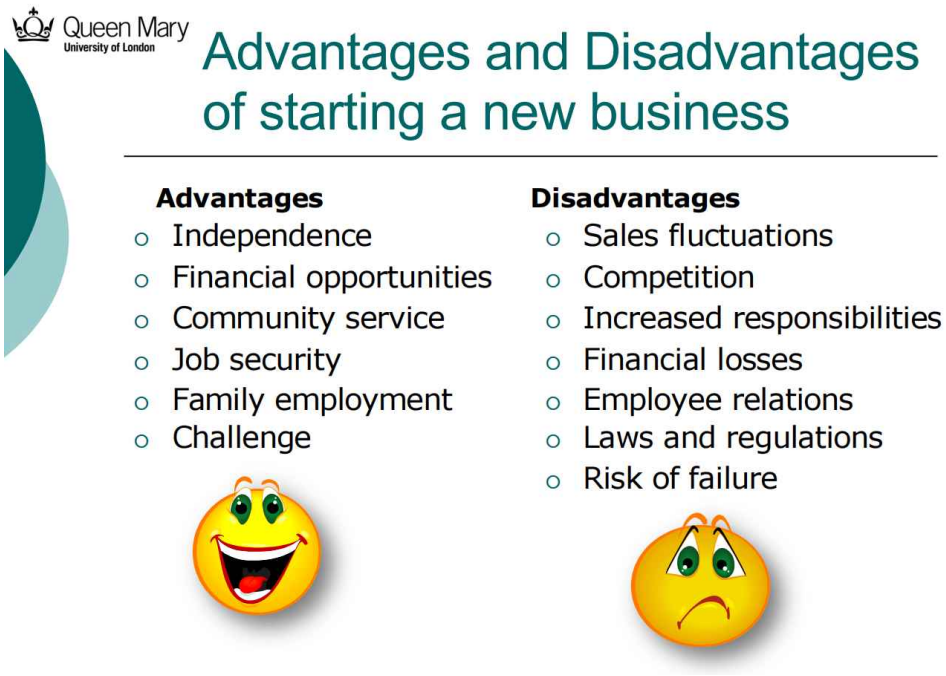
¡ Hard work, making most decisions on your own

¡ Considerable risks involved

¡ Costly to raise finance

¡ No economies of scale

Pros and Cons



Who are your customers?

¡ End-users, i.e. consumers

l i.e. you are B2C (business to consumer)

¡ Business that has its own customers

l i.e. you are B2B (business to business)

¡ You **must** know who your potential customers are

How a hi-tech start-up comes together

The Ingredients:

¡ Technology

¡ Intellectual Property (Protection)

¡ The Vision (Where can the technology lead? Is there a market for it)

¡ Personnel

l Scientific leaders

l Business managers

l Fund Raisers

l Scientists/Engineers

¡ Money

The stages for creating a hi-tech start-up

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i. **Assess** the opportunities - generate, evaluate &

refine the business concept

ii. **Develop** the business plan and decide on the

structure of the venture

iii. **Acquire** the necessary resources and funding

iv. **Grow** and harvest the venture

Raising capital

¡ **Why**?

l Growth

l Maintenance of operations

l Survival

¡ **How**?

l Re-mortgage or loan

l Venture Capitalists or Angels

l Grants

l Family and friends

Fund-raising Stages

¡ Elevator pitch/executive summary

¡ Business plan, presentation, management team

¡ Business survey

¡ Due diligence

¡ Valuation, ownership, control, legal issues

¡ FUNDING

Business Plan

1. Describes:

   l The **market** you plan to enter

   l The unique and compelling **features** of your contribution to this market

   l **Intellectual property**

   l **Business model** and **Financials**

   l Team and Company development and **strategy**

   l **Investment** proposition

   l **Exit** strategy

2. A business plan:

   ¡ provides a **formal agreement** between the founders of a company about the direction to be taken

   ¡ can reduce **self-delusion** amongst the founders

   ¡ defines **responsibilities** & rewards

   ¡ helps to translate **abstract** goals into **explicit** operational needs

3. Key Element:

   ¡ Title Page and Table of Contents

   ¡ Executive Summary

   ¡ Vision and Mission Statement

   ¡ Company History

   ¡ Business and Industry Profile

   ¡ Business Strategy

   ¡ Description of Products/Services

   ¡ Marketing Strategy

      l Document market claims

      l Show customer interest

   ¡ Competitor Analysis

   ¡ Description of Management Team

   ¡ Plan of Operation

¡ Projected Financial Statements

¡ Loan or Investment Proposal

Causes of business failure

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l lack of **skills**

l **sales** problem

l **financial** control

l lack of **funds**

l high cost of **finance**

l insolvent **customers**

l overtrading **development**

l **marketing** issues

l red tape (**bureaucracy**)

Topic 19

What is Cybersecurity?

¡ Cybersecurity is the application of technologies, processes and controls to protect systems, networks, programs, devices and data from cyber attacks

¡ It aims to reduce the risk of cyber attacks, and protect against the unauthorised exploitation of systems, networks and technologies

¡ Increasingly implemented through laws and regulations

¡ Three distinct legal elements: information security, privacy and data protection, and cybercrime

Information Security

¡ Seeks to protect all information assets, whether in hard copy or in digital form

¡ Information is one of the most valuable assets

¡ Good business practice

¡ Digital revolution changed how people communicate and conduct business

¡ New possibilities & challenges

Security x Privacy

¡ Information security and privacy are closely related, but distinct concepts

¡ Privacy is an individual' s right to control the use and disclosure of their own personal information

¡ Information security is the process used to keep data private

¡ Security is the process; privacy is the result

Mitigating risks to the trustworthiness of information of corporations and governments by the:

¡ Development of strategies and

¡ Implementation of technologies and procedures in order to preserve its

l confidentiality,

l integrity, and

l availability

appropriate to the time and circumstances

Key Concept

¡ Risk management as means to justify information security laws

= process of listing the all the relevant factors and taking steps to control them where possible

¡ There are four main concepts:

l Vulnerabilities

l Threats

l Risks

l Safeguards

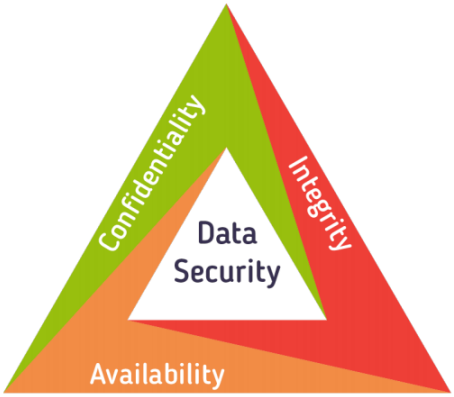
1)Processes, procedures and infrastructure to preserve:

– confidentiality

– integrity, and

– availability of information

2)Diagram



Confidential

¡ Confidentiality means that only people with the right permission can access and use information

¡ Protecting information from unauthorised access at all stages of its life cycle

¡ Information must be created, used, stored, transmitted, and destroyed in ways that protect its confidentiality

¡ Ensuring confidentiality – encryption, access controls

¡ Compromising confidentiality – (intentional) shoulder surfing, social engineering; (accidental) publication

¡ It may result in identity theft, threats to the public safety

Integrity

¡ Integrity means that information systems and their data are accurate

¡ Changes cannot be made to data without appropriate permissions

¡ Ensuring integrity – controls ensuring the correct entry of information, authorization, antivirus

¡ Compromising integrity – (intentional) employee or external attacks; (accidental) employee error

Authentication

¡ Specific to integrity and confidentiality considerations

¡ Authentication is the process of validating the identity of a registered user or process before enabling access to protected networks and systems.

¡ Analogue

l signatures, handwriting, in person attestation, witnesses, notary

¡ Digital

l username and password, digital signatures, fingerprints or face recognition

Availability

¡ Availability is the security goal of making sure information systems are reliable

¡ Data is accessible

¡ Individuals with proper permission can use systems and retrieve data in a dependable and timely manner

¡ Ensuring availability – recovery plans, backup systems

¡ Compromising availability – (intentional) denial of service (DoS) attack, (accidental) outage

Vulnerabilities

= weakness or flaw in the information system that can =be exploited

l Construction, design mistake

l Flaws in how internal safeguards are used/not used, based on:

i. People
ii. Process
iii. Facility
iv. Technology

¡ People

l separation of duties principle

l Two or more people need to split critical task functions

¡ Process

l flaws in organization’ s procedures

l missing step in a checklist /no checklist

l failure to apply hardware and software patches

= patch is a software/code that updates a program to address security problems

¡ Facility

l flaws in physical infrastructure

l fences, locks, CCTV cameras

¡ Technology

l design flaws

l unpatched applications, improperly configured equipment

Successful attacks take place when vulnerability is exploited

Threats

= anything that can cause harm to an information system – successful exploits of vulnerabilities

¡ Relationship between a vulnerability and a threat

l An organization does not have sufficient controls to prevent an employee from deleting critical computer files (lack of controls – vulnerability). An employee could delete files by mistake (employee – source of threat) (deleting critical files – threat). If the files are deleted, successful exploit of the vulnerability has taken place. If the file is not recoverable, the incident harms the organizations and its security. Availability is compromised.

i. Human

l Internal and external, includes well-meaning employees and external attackers

ii. Natural

l uncontrollable events (fire, flood)

iii. Technology and operations

l operate inside information systems (malicious code, hardware and software failures)

iv. Physical and environmental – lack of physical security

l Accidental or intentional

l Internal or external attackers

Risks

= a likelihood that a threat will exploit vulnerability and cause harm, where the harm is the impact to organization

Risk = vulnerability + threat

¡ Risks can occur at any layer of the information system:

l At the physical hardware or device layer, e.g. when a flood renders servers stored in a

basement unavailable;

l At the various software layers, e.g. when hackers exploit a vulnerability in software;

l At the network layer, e.g. when a hacker intercepts data packets as they pass through the

network from sender, via routers, to receiver; or,

l At the user layer, e.g. through ‘social engineering’, such as convincing users to share

their passwords through ‘phishing’ emails

¡ Risk analysis and management to classify and respond to risks

¡ Probability a threat will exploit a vulnerability – high, medium, low

¡ Information security impact – loss of confidentiality, integrity and availability

¡ Other impacts – loss of life, productivity or profit, property and reputation

¡ Assessment of impact – address risks that have large impact on information security

¡ Types of responses: risk avoidance, risk mitigation, risk transfer, risk acceptance

Safeguards

= safeguard reduces the harm posed by information security vulnerabilities or threats

Administrative – actions and rules implemented to protect information (need to know rule)

Technical – logical rules that state how systems will operate (least privilege rule)

Physical – actions to protect actual physical resources

Information Security Management

¡ To effectively assess the security needs of an organization and to evaluate and choose various security products and policies

¡ Categorize information

¡ Identify legal obligations

¡ Assess vulnerabilities, threats and risks

¡ Safeguards

¡ There are a number of challenges

¡ Development of standards

Topic 7 Innovation

To **renew**, to **make new** or to **alter,** new way of doing things

Innovation characteristics:

l Process from idea generation to commercialization

l The adoption of change,

l Radical change in traditional ways vs. incremental change,

l New device or something new to society

Difference between

An **invention** can still become an **innovation** even if it is **unsuccessful**, provided it made it to the marketplace

Innovation and Product Development

¡ An innovation is successful implementation of a **new** or significantly **improved product** (good or service), or **process**, a **new marketing method**, or a **new organisational method** in business practices,workplace organisation or external relations.

¡ A creative process in which **two or more existing things are combined** in some novel way to produce unique new thing.

¡ Innovation is the management of all the activities involved in the process of idea generation, technology development, manufacturing and marketing of a new (or improved) product or manufacturing process or

equipment.

Type of innovation

Product Innovation	The development of a new or improved product
Process innovation	The development of a new manufacturing process
Organisational innovation	A new venture division, a new internal communication system, introduction of a new accounting procedure
Management innovation	TQM (Total Quality Management) systems. BPR (business process re-engineering)
Production innovation	Quality circles, just-in-time (JIT) manufacturing system, new production planning software
Commercial/marketing innovation	New financing arrangements, new sales approach e.g. direct marketing, e-commerce
Service innovation	Telephone financial services or on-line services e.g. customer support

Product Innovation

In most cases we think of innovation relating to **actual products**

l New technologies

l Leading edge products

l Gadgets, etc

Advantages

of launching new or improved products on to the market includes:

l Increased market share

l Public relations

l Enhanced reputation as an innovative company

l Opportunity to build early customer loyalty

l Added value

l Higher prices and profitability

l Competitive advantage

can lead to **process**, **management** and/or **organisational** innovation

Process Innovation

Producing、Delivering、new product

Advantages are

l Reduced **costs**

l Improved **quality**

l More responsive **customer service**

l Greater **flexibility**

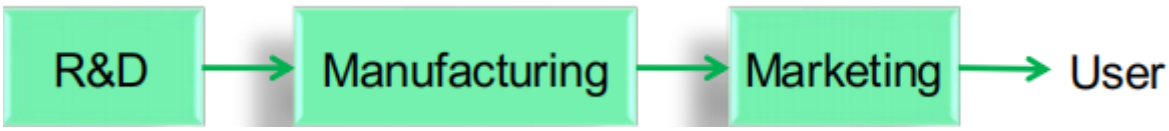
Disadvantages

l Loss of **jobs** – especially if work is outsourced

l Need for **re-training** of workers

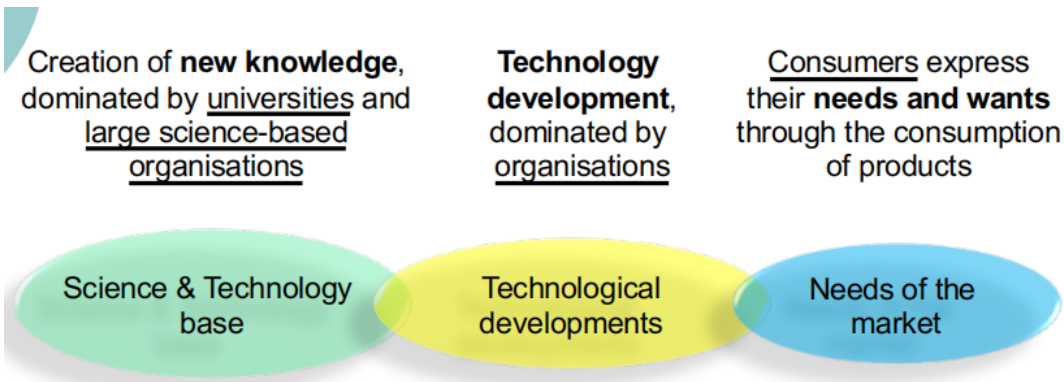
Models of innovation

source：**Serendipity、Technology push**

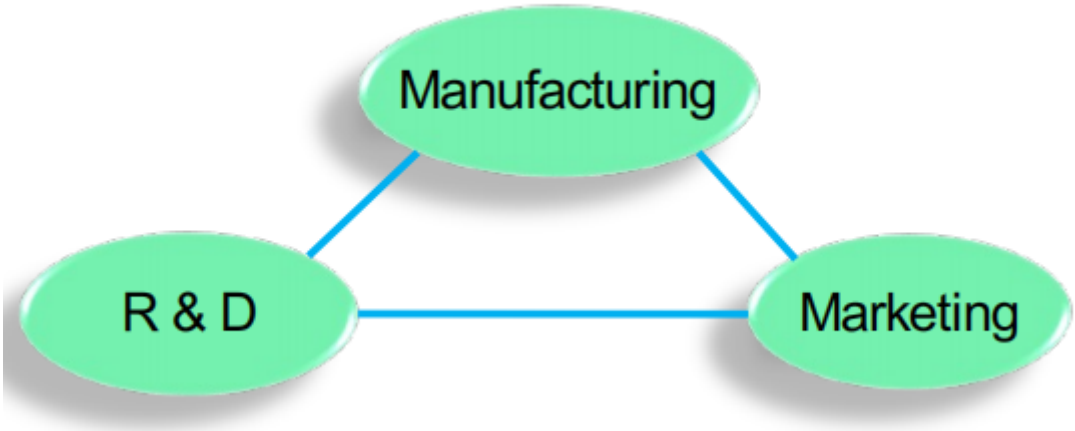


market pull：

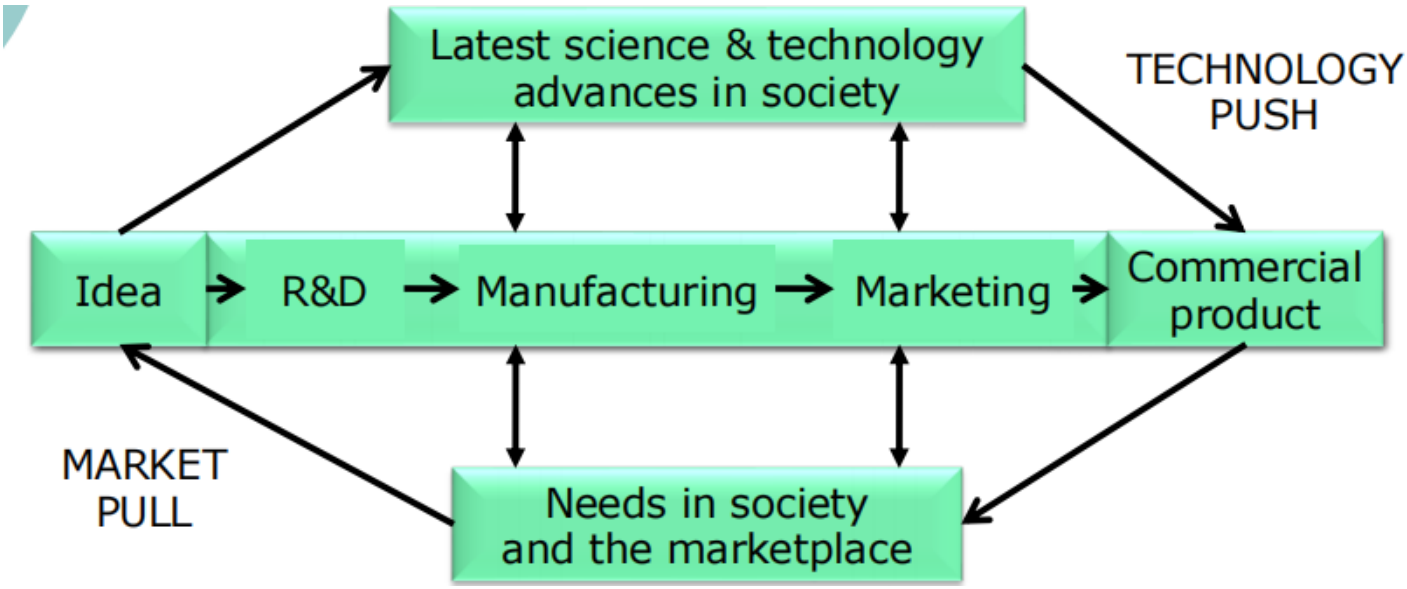
university & industry liaison:



simultaneous coupling:



interactive:



What makes an innovative company

investment : Growth orientation

Vigilance:

¡ be aware of what markets and competitors are doing

¡ **SWOT**

¡ Identifying **threats**, **opportunities**

¡ encouraged in **all departments** at all levels

¡ **R&D** team University liaison

¡ **monitor** competition and opportunities

commitment to technology:

¡ **Long-term** commitment is needed, and appropriate resources must be financed

¡ There is not always an immediate return on **investment**

¡ **Creativity** is only seen in a climate of commitment to innovation

acceptance of risks:

¡ Not all investment in innovation will be realised in **turnover** or **profit**

¡ Not all inventions will become **innovations**

¡ **Risk assessment** strategies and balanced product portfolios are crucial to success

¡ The business **strategy** must consider risk and commitment

cross-functional cooperation:

¡ Reduction/removal of inter-departmental **conflict**

¡ The biggest conflict is often between **marketing** and **R&D** functions

Receptivity:

¡ The ability to accept technologies from **outside** of the company and use them to create new innovations

¡ It is very difficult to create everything 'in-house'

¡ Strength to identify areas of **focus**

¡ **Joint activities** allow knowledge sharing

Slack:

¡ Allowing individuals time to **think** and investigate new technologies etc.

¡ Some **flexibility** on milestones

¡ **Encouragement** for group discussions, attending conferences etc.

Adaptability:

¡ Ability for companies to **adapt** as required

¡ Innovative products can lead to **changes** in process

¡ Need to encourage a **climate** that accepts change

¡ Can be easier in **smaller companies** with less processes.

diverse range of skills:

¡ Ensuring that a company has a **work force** with a diverse range of skills

¡ Important to have employees with the appropriate **industry knowledge**

¡ Good **management** knowledge and experience

¡ Strong overall team with the correct **mix** of skills and experience

¡ Innovation ecosystems rely on a wide range of skills,

Topic 8 – Digital Transformation and Digital Products

Digitisation vs. Digitalisation vs. Digital Transformation

Digitisation

is the process of converting information from analog to digital.

Digitalisation

is the process of using digitised information to make established ways of working simpler and more efficient.

Digital transformation

is the process of using digital technologies to create new — or modify existing — business processes, culture, and customer experiences to meet changing business and market requirements

Digital Transformation

4 types:

☒ **Process Transformation** – aims to process such as data, analytics, AI, and any process that can work towards lowering costs and driving operational efficiency in the business.

☒ **Business Model Transformation** – aims to make fundamental changes in how a business or organization runs which can include personnel, processes, and technology.

☒ **Domain Transformation** - This area offers a great opportunity to move into a new domain or area that a business may not have explored before by acquiring new technologies.

☒ **Cultural/Organizational Transformation** - This is about redefining mindsets, processes, capabilities and skills for a digital world. It’ s about driving digital transformation forward through growth initiatives that are grounded in a new culture and way of thinking.

trends：

☒ **5G**

☒ **Zero-trust security**

☒ **Hyperautomation**

☒ **Predictive analytics**.

☒ **Customer Relationship Management**

☒ **Generative AI**

Digital Product

☒ **no physical** form, exist only in the digital realm,

☒ **intangible** items delivered **electronically**,

☒ anything that can be **downloaded** and **used digitally** can be considered a digital product,

☒ sold online or through brick-and-mortar retailers,

☒ can be easily updated or modified to keep up with changing technology and trends because they're intangible,

☒ often come with a license that allows a customer to use them in unlimited ways.

most products and experiences they are part of are **hybrid**

A digital copy-cat of a physical product is usually not the best way to go

Advantage：

☒ Customer happiness is how you win in business. Modern customer expectations are being driven by largely digital technology and digital innovations.

☒ Low investment, (potentially) high returns

☒ More profitable than physical goods

☒ No inventory, shipping or rent hassle

☒ Automated delivery for passive income

☒ Serve a niche at scale

☒ Digital products offer unique ways to communicate directly with the customers.

6 steps of Digital Project Development

☒ **Phase 1. Discovery**

☒ **Phase 2. Ideate**

☒ **Phase 3. Test**

☒ **Phase 4. Execute**

☒ **Phase 5. Launch**

☒ **Phase 6. Grow**

Digital Project Manager：

brings together many facets of a successful digital product

Project Team for Digital Products：

Developers or **engineers**

Experts in **customer or user experience**

Sales and marketing experts

mistakes to avoid

☒ Having the wrong data

☒ Resistance from staff

☒ Underestimate costs

☒ A lack of commitment

☒ A lack of skills